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Essential Mathematics for Datascience

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Lecture 2



- Consider a vector space V and a set of vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k \in V$.
- Then every $\mathbf{v} \in V$ that is of the form $\mathbf{v} = \lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2 + \dots + \lambda_k \mathbf{x}_k$ is a linear combination of the vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$.
- Note that the $\mathbf{0}$ can be written trivially as a linear combination of the given k vectors $\mathbf{x}_i$ s. We are however interested in non-trivial linear combinations of vectors to get the $\mathbf{0}$ -vector.



- Consider a vector space V and k vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k \in V$. Then if there is a non-trivial linear combination of the given vectors such that $0 = \sum_{i=1}^k \lambda_i \mathbf{x}_i$ where at least one of the λ_i s is non-zero, then the vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ are linearly dependent.
- Put another way, if the only way we can combine vectors to get the $\mathbf{0}$ -vector is by letting all the λ_i be zero, then the vectors are linearly independent.



- k vectors in a vector space are either linearly independent or linearly dependent. There is no third option.
- If at least one of the vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ is the 0-vector, then the vectors are linearly dependent. Why? Apply the definition to see that this is the case. We can choose a non-zero λ for the 0-vector and zero λ s for all the other vectors to get the linear combination to be zero.
- If all the vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ are non-zero, then the vectors are linearly dependent if and only if one of them is a linear combination of the others.

Algorithm for checking Linear Independence



- A practical way of checking whether a bunch of vectors are linearly independent is to fill out the columns of a matrix with the given vectors and then convert A into its row-echelon form
- The pivot columns indicate all the vectors that are linearly independent, and they are all on the left.
- The non-pivot columns can be expressed as a linear combination of the pivot columns

Example



■ Consider the matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 4 \end{bmatrix}$

■ Its row-echelon form is

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -2 \end{bmatrix}$$

■ The pivot columns are the first and the third column and we see that the second column which is a non-pivot column can be expressed as a linear combination of the pivot columns to its left, which is just twice the first column.



An Abelian Group is a set G and an operation $\otimes : G \times G \rightarrow G$ defined on G . Then $(G, \otimes)$ is called an Abelian group if the following properties hold:

- a) Closure of G under $\otimes$: $\forall x, y \in G, x \otimes y \in G$
- b) Associativity: $\forall x, y, z \in G, (x \otimes y) \otimes z = x \otimes (y \otimes z)$
- c) Neutral or Identity element: $\exists e \in G, \forall x \in G, x \otimes e = x$
- d) Inverse element: $\forall x \in G, \exists y \in G, x \otimes y = y \otimes x = e$
- e) Commutative $\forall x, y \in G, x \otimes y = y \otimes x$ then $(G, \otimes)$ is an Abelian group

- a) $(\mathbb{Z}, +)$ is an Abelian group. We can see that all the aforementioned properties hold.
- b) What about $(\mathbb{N}_0, +)$? This is not a group since there is no inverse element for an arbitrary element in it.
- c) $(\mathbb{Z}, \cdot)$ where $\mathbb{Z}$ is the set of integers, and $\cdot$ is product. It has the identity element but there exist elements in it that don't have an inverse.



Consider sets that are just like groups in terms of its properties with respect to an inner operation $+$, and also have an outer operation called $.$ which denotes the multiplication of a vector $x \in G$ by a scalar $\lambda \in \mathbb{R}$.

A real-valued vector space is a set $V = (\mathcal{V}, +, .)$ with operations $+, .$ such that $+: \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$ and $.: R \times \mathcal{V} \rightarrow \mathcal{V}$ where $(\mathcal{V}, +)$ is an Abelian group, and the following properties hold:

- a) Distributivity
- b) associativity with respect to the outer operation.
- c) there exists a neutral or identity element with respect to the outer operation.



Let us look at the properties of a Vector Space more carefully:

- a) Distributivity: $\forall \lambda \in \mathbb{R}, \mathbf{x}, \mathbf{y} \in \mathcal{V}, \lambda.(\mathbf{x} + \mathbf{y}) = \lambda.\mathbf{x} + \lambda.\mathbf{y}$ and $\forall \lambda, \psi \in \mathbb{R}, (\lambda + \psi).\mathbf{x} = \lambda.\mathbf{x} + \psi.\mathbf{x}$
- b) Associativity with respect to the outer operation:
 $\forall \lambda, \psi \in \mathbb{R}, \mathbf{x} \in \mathcal{V}, \lambda.(\psi.\mathbf{x}) = (\lambda\psi).\mathbf{x}$
- c) Neutral element with respect to the outer operation: $\forall \mathbf{x} \in \mathcal{V}, 1.\mathbf{x} = \mathbf{x}$
- d) The elements $\mathbf{x} \in \mathcal{V}$ are called vectors.
- e) The neutral element with respect to $(\mathcal{V}, +)$ is the zero vector $[0, 0, \dots, 0]^T$ and the inner operation is called vector addition.



Consider some examples of vector spaces, for example $\mathcal{V} = \mathbb{R}^n$

We can define addition:

$$\mathbf{x} + \mathbf{y} = (x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$$

Multiplication by scalars:

$$\lambda \mathbf{x} = \lambda(x_1, x_2, \dots, x_n) = (\lambda x_1, \lambda x_2, \dots, \lambda x_n), \forall \lambda \in R, \mathbf{x} \in \mathbb{R}^n.$$



What we call a vector need not be the standard column vector that we are accustomed to treating as a vector. We can think of $m \times n$ matrices as vectors and create a vector space out of them. Thus $\mathcal{V} = \mathbb{R}^{m \times n}$ with addition and multiplication defined as below:

$$\text{Addition } \mathbf{A} + \mathbf{B} = \begin{bmatrix} a_{11} + b_{11} & \dots & a_{1n} + b_{1n} \\ \vdots & & \vdots \\ a_{n1} + b_{n1} & \dots & a_{nn} + b_{nn} \end{bmatrix} \text{ is defined}$$

element-wise for two matrices $A, B \in \mathcal{V}$.

$$\text{Multiplication by scalars: } \lambda \mathbf{A} = \begin{bmatrix} \lambda a_{11} & \dots & \lambda a_{1n} \\ \vdots & & \vdots \\ \lambda a_{n1} & \dots & \lambda a_{nn} \end{bmatrix}$$



- Let $V = (\mathcal{V}, +, \cdot)$ be a vector space and let $\mathcal{U} \subseteq \mathcal{V}, \mathcal{U} \neq \Phi$. then $U = (\mathcal{U}, +, \cdot)$ is called a vector subspace of V if U is a vector space with the vector space operations $+$ and $\cdot$ restricted to $\mathcal{U} \times \mathcal{U}$ and $R \times \mathcal{U}$.
- We use the notation $U \subseteq V$ to denote that U is a vector subspace of V .
- If $U \subseteq V$ and V is a vector space, then U naturally inherits many properties directly from V because they hold for all $\mathbf{x} \in V$, and in particular for all $\mathbf{x} \in U \subseteq V$. These peroperties include the Abelian group property, associativity, distributivity and the neutral element.

How do we show if $(\mathcal{U}, +, \cdot)$ is subspace of V ?

We need to show that

- a) $\mathcal{U} \neq \Phi$
- b) Closure of $\mathcal{U}$ with respect to the outer and inner operations,
i.e $\forall \lambda \in R, \forall \mathbf{x} \in \mathcal{U}, \lambda \mathbf{x} \in \mathcal{U}$ and $\forall \mathbf{x}, \mathbf{y} \in \mathcal{U}, \mathbf{x} + \mathbf{y} \in \mathcal{U}$.

- Let $\mathcal{V} = \mathbb{R}^2$ and $\mathcal{U}$ be the y -axis. Is $U = (\mathcal{U}, +, \cdot)$ a subspace of $V = (\mathcal{V}, +, \cdot)$? Answer: Yes, because the addition of any two vectors on the y -axis remains on the y -axis so closure with respect to the addition operation is satisfied. Also any vector on the y -axis when scaled by any real number (including 0) will yield a vector on y -axis.
- $U = \text{span}(\mathbf{x}_1, \dots, \mathbf{x}_m) \subseteq \mathbb{R}^n$ is a subspace



- What about the subset of $\mathbb{R}^2$ that represents a square around the origin , i.e $-1 \leq x \leq 1, -1 \leq y \leq 1$? This is again not a subspace.
- This subspace is of particular interest to us called the nullspace of a matrix, i.e the set of solutions to a linear system of equations $Ax = 0$. Why is this a vector subspace?

Span of a set of vectors



- Consider a vector space $V = (\mathcal{V}, +, \cdot)$ and a set of vectors $\mathcal{A} = \{x_1, x_2, \dots, c_k\} \subseteq V$.
- The set of all linear combinations of the vectors in $\mathcal{A}$ is known as the **span** of $\mathcal{A}$.
- If $\mathcal{A}$ spans the vector space V we write $V = \text{span}[\mathcal{A}]$.